

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Practical Lab Exam 1

Practical Lab Exam 1

Class: Class 10

Assessment Type: Practical Lab Exam 1

Total Marks: 20

Duration: 45 minutes

Topics Covered: HTML/CSS and Python Programming

ICT Teaching Resources Portal

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Lab Tasks (15 marks)

Task 1: HTML/CSS - Personal Portfolio Website (5 marks)

Create a complete HTML/CSS webpage for a **Personal Portfolio** that includes:

- Hero section with name, photo placeholder, and tagline
- About section with bio (minimum 3 sentences)
- Skills section displaying at least 5 skills with visual progress bars (HTML/CSS only)
- Projects section with at least 3 project cards (image placeholder, title, description)
- Contact form with name, email, subject, and message fields
- Responsive design using media queries (mobile-friendly)
- Professional styling with consistent colour scheme

Task 2: Python - Student Records Manager (5 marks)

Write a Python program that:

- Creates a list of dictionaries to store student records (name, roll, marks in 3 subjects)
- Implements a menu-driven interface with options:
 1. Add new student
 2. Display all students
 3. Search student by roll number
 4. Display students sorted by total marks
 5. Display class statistics (average, highest, lowest)
 6. Exit
- Uses proper functions for each operation
- Includes input validation
- Displays results in a formatted table

Task 3: Python - File Handling with Exception (5 marks)

Write a Python program that:

- Reads a text file (user-specified filename)
- Counts total words, lines, characters, and unique words
- Finds the 5 most frequently used words
- Writes a summary report to a new file
- Uses try-except blocks for:
 - FileNotFoundError
 - PermissionError
 - General Exception
- Displays proper error messages for each exception type

Viva Voce (5 marks)

1. What is the difference between `display: none` and `visibility: hidden` in CSS? When would you use each? (1 mark)
2. Explain the 盒模型 (box model) in CSS. What happens when you set `box-sizing: border-box`? (1 mark)
3. Why is `with open()` preferred over `open()` and `close()` in Python file handling? (1 mark)
4. What is the difference between `read()`, `readline()`, and `readlines()` methods? When would you use each? (1 mark)
5. How does Python handle exceptions? What is the purpose of `else` and `finally` blocks in exception handling? (1 mark)